Trainings ▾

General Information

Cummins Inc. recommends that the engine be maintained according to the maintenance schedule below.

If the engine is operating in ambient temperatures consistently below -18°C [0°F] or above 38°C [100°F], perform maintenance at shorter intervals. Shorter maintenance intervals are also required if the engine is operated in a dusty environment or if frequent stops are made. Contact a Cummins® Authorized Repair Location for recommended intervals.

If your engine is equipped with a component or an accessory **not** manufactured by Cummins Inc., reference the component manufacturer's maintenance recommendations.

Most of the maintenance operations described in this manual can be performed with common hand tools (metric and SAE wrenches, sockets, and screwdrivers).

Below is the maintenance schedule for the K19 engine. Perform maintenance at whichever interval occurs first. At each scheduled maintenance interval, perform all previous maintenance checks that are due for scheduled maintenance.

As Needed or Every 4000 Hours

- Centinel™ Make-up Tank - Check

Every 25 Hours

- Centinel™ System - Check

Maintenance Procedures at Daily Interval (Section 3)

- Engine Operation Report - Check
- Fuel-Water Seperator - Drain
- Lubricating Oil Level - Check
- Coolant Level - Check
- Drive Belts - Check
- Air Cleaner Precleaner - Check
- Sea Water Strainer - Clean

Maintenance Procedures at Weekly Intervals (Section 4)

- Air Intake Piping - Check
- Air Cleaner Restriction Indicator - Check
- Air Tanks and Reservoirs - Drain

Maintenance Procedures at 250 Hours or 6 Months (Section 5)

- Lubricating Oil and Filters[1] - Change
- Lubricating Oil Analysis - Check
- Fuel Filter (Spin-On Type) - Change
- Coolant Filter - Replace
- Air Compressor Air Cleaner Element - Change
- Crankcase Breather Tube - Check
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Measure
- Drive Belt, Cooling Fan - Check
- Fan, Cooling - Inspect for Reuse

Maintenance Procedures at 1500 Hours or 1 Year (Section 6)

- Engine Steam Cleaning - Clean
- Overhead Set (OBC)[2] - Adjust
- Overhead Set (Travel Method)[2] - Adjust
- Fan Drive Idler Arm Assembly - Check
- Engine Support Bracket, Front - Check
- Radiator Hoses - Inspect for Reuse
- Batteries - Check
- Engine Mounts - Check
- Crankshaft - Measure
- Zinc Anode - Inspect for Reuse
- Crankcase Breather (External) - Inspect for Reuse

Maintenance Procedures at 6000 Hours or 2 Years (Section 7)

- Fuel Pump - Inspect for Reuse
- Injector - Inspect for Reuse
- Turbocharger - Check
- Vibration Damper, Rubber (if installed) - Check
- Vibration Damper, Viscous (if installed) - Check
- Air Compressor Discharge Lines - Check
- Air Compressor Unloader and Valve Assembly - Inspect for Reuse
- Fan Hub, Belt Driven (if installed) - Check
- Fan Hub, Gear Driven (if installed) - Check
- Water Pump - Check
- Cooling System - Flush[4]
- Fan Drive Idler Pulley Assembly - Inspect for Reuse

Other Maintenance

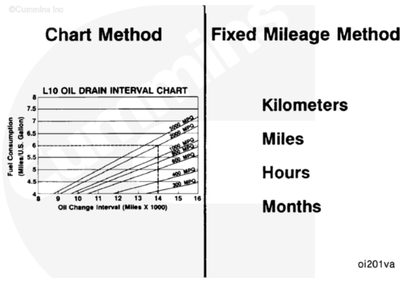
- Alternator[3] - Check
- Generator[3]- Check
- Starter[3]- Check
- Air Compressor (Non-Cummins)[3]- Check
- Electrical connections[3]- Check
- Batteries[3]- Check
- Fan Shaft Bearings[3] - Check
- Clutch or marine gear[3] - Check

- Refrigerant compressor[3] - Check
- Hydraulic governor[3] - Check

1. Reference the Oil Drain Intervals for an alternate method of determining safe oil drain intervals.
2. Cummins Inc. has found that engines in most applications will **not** experience significant valve or injector train wear after an initial adjustment is made at 1500 hours. After this adjustment, it is recommended that the valves and injectors **not** be adjusted again previous to injector calibration at the 6000 hours or 2 years interval. Because injector train hardware is typically mixed between cylinders during injector replacement, it is recommended to adjust valves and injectors 1500 hours after all injector replacements.
3. On these components, follow the manufacturer's recommended maintenance procedure.
4. Flush in this instance. The cooling system requires the system to be drained/flushed/filled.

Oil Drain Intervals

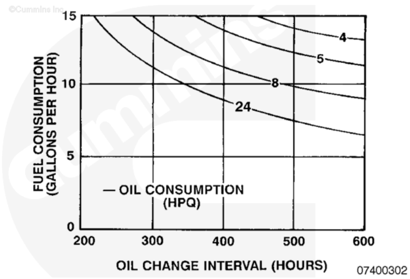
Note : Do not extend oil and filter change intervals beyond 250 hours or 6 months (except generator drives) unless the Chart Method is used. On generator drives, the intervals are 250 hours or 12 months, whichever occurs first. Refer to the charts below. Extended oil and filter change intervals will decrease engine life due to factors such as corrosion, deposits, and wear.



There are two recommended methods used to determine the proper oil and filter change interval:

- Chart Method (based on known fuel and oil consumption rates).
- Fixed Mileage Method (based on fixed kilometers, miles, hours, or months; whichever occurs first).

The Chart Method is recommended to provide the lowest total cost of operation while still protecting the engine.



Use the Chart Method with the required information listed below to determine the correct oil and filter change interval for your engine:

- Fuel consumption rate
- Oil consumption rate
- Total system capacity.

Determine fuel and oil consumption rates:

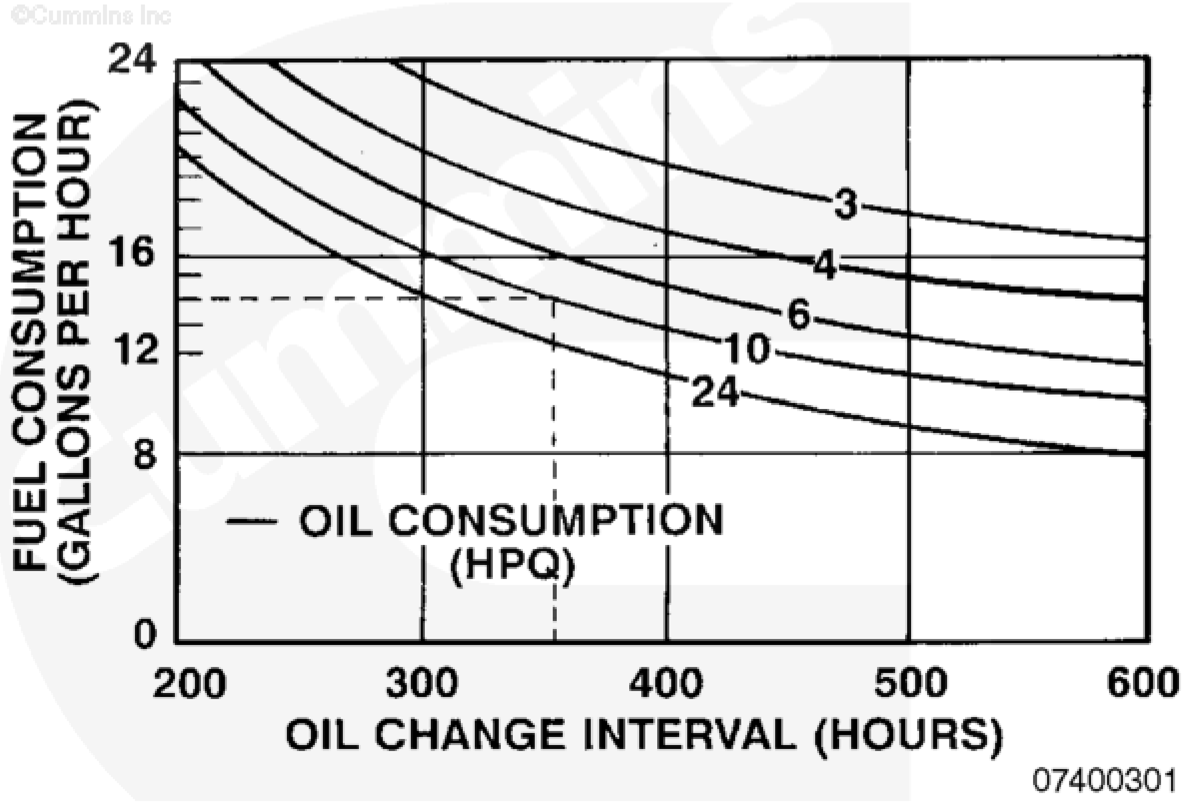
- To use the Chart Method effectively, accurate fuel and oil consumption records **must** be kept and maintained.
- As oil and fuel consumption rates change as a result of a change in operation or duty cycle of a particular engine, the oil change interval established by the Chart Method **must** be re-evaluated based on the change in oil and (or) fuel consumption.
- Use the following procedure for oil pan capacities. Refer to Procedure 018-017 in Section V. (/qs3/pubsys2/xml/en/procedures/18/18-018-017.html)
- Total lubricating oil system capacity can be determined by adding the high level of the oil in the oil pan plus the capacity of the full flow and by-pass oil filters. Use the following procedure for oil filter capacities. Refer to Procedure 018-017 in Section V. (/qs3/pubsys2/xml/en/procedures/18/18-018-017.html)

As an example: The KTA19 engine has oil pan, Part Number 3200709, and utilizes the standard full-flow filter head (2x LF670 filters) and one spin-on by-pass filter (LF777).

Total capacity equals:	
Oil Pan	47.32 liters [12.5 U.S. gal]
Two - LF670 Filters	5.3 liters [1.4 U.S. gal]
One - LF777 Filter	2.3 liters [.6 U.S. gal]
Total System Capacity	54.89 liters [14.5 U.S. gal]

- Round this capacity to the nearest whole U.S. gallon (15 U.S. gallons) and select the appropriate chart.

For our example, assume the average fuel consumption equals 14 U.S. gallons per hour and the average oil consumption equals 10 hours per U.S. quart.



To read the chart:

Select the chart entitled 15 U.S. gallons.

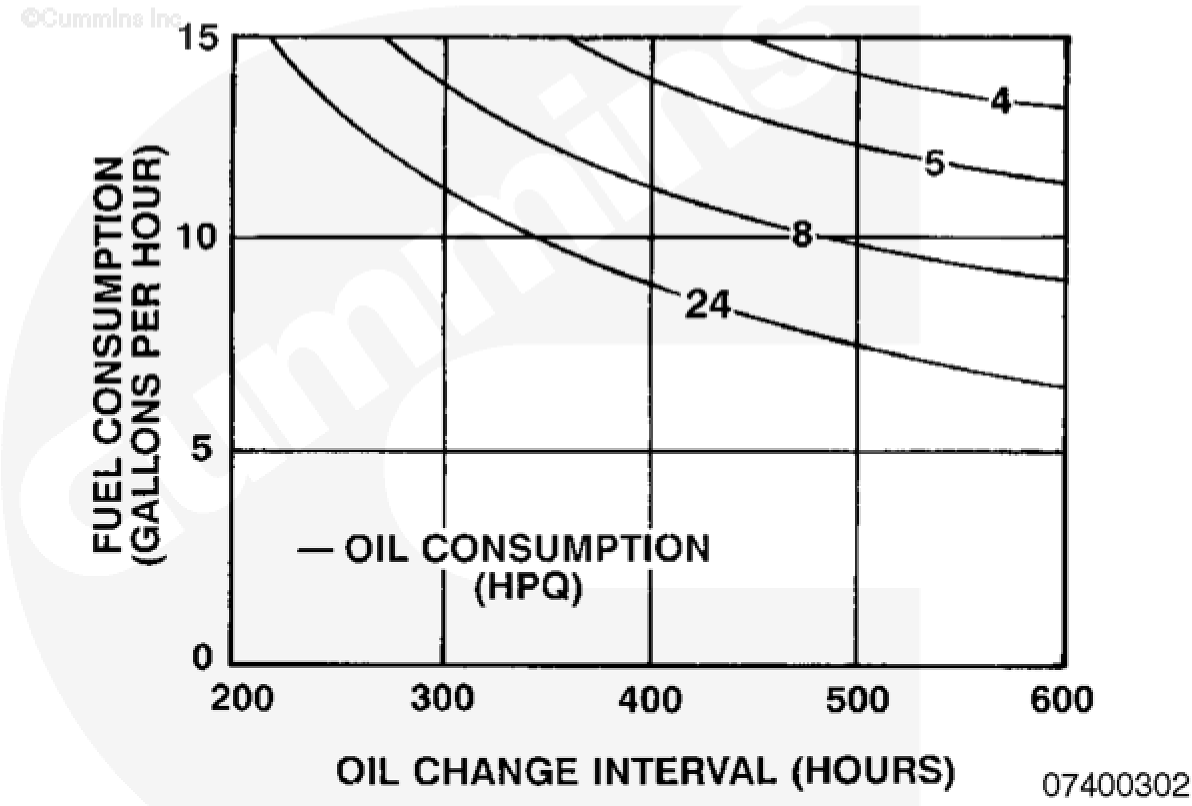
The left vertical axis of the chart represents fuel consumption in U.S. gallons per hour.

Determine the location of 14.0 gallons on the left vertical axis and draw a line from left to right across the chart, parallel with the bottom of the chart, until it intersects with the curve marked 10 (10 hours per quart).

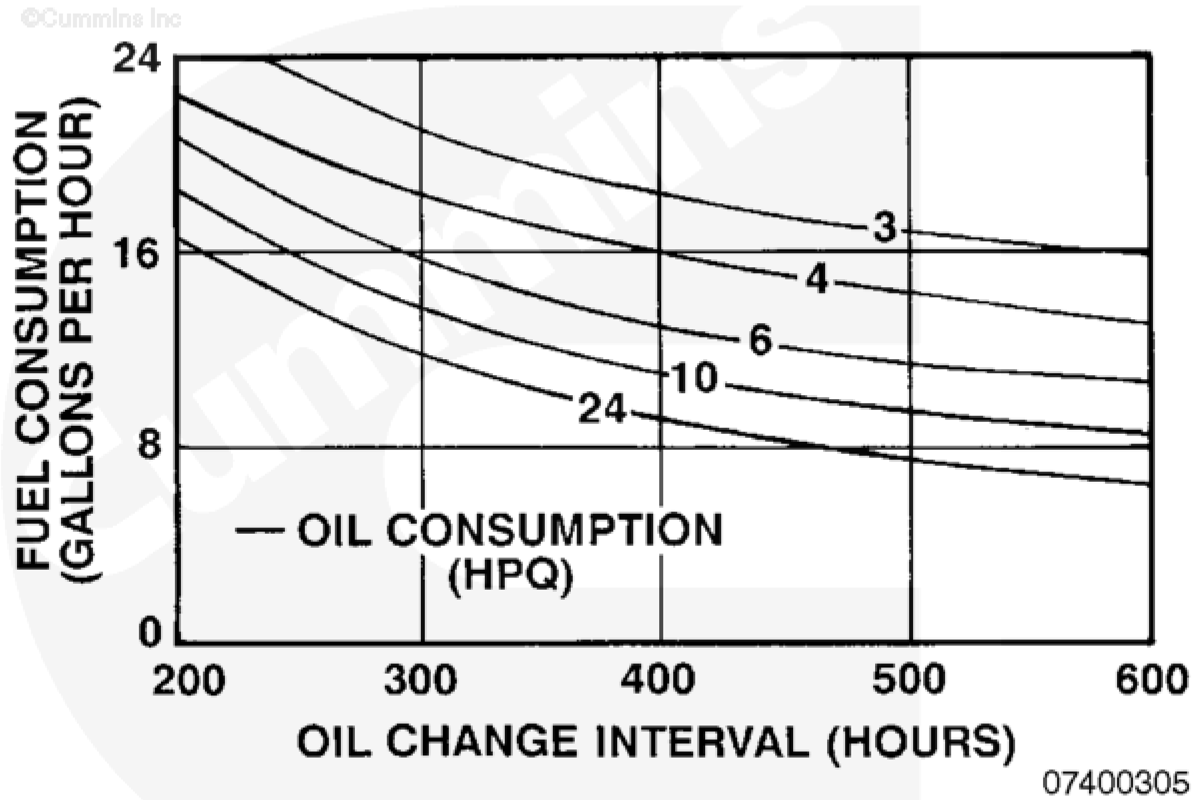
From the intersection point on the curve 10, draw a line perpendicular to the bottom of the chart. The number across the bottom of the chart represents the oil change interval in hours. In this case, the total oil capacity, oil consumption, and fuel consumption of this engine indicates that an oil change interval of 355 hours is recommended.

The charts that follow will allow oil change intervals to be calculated for the total lubricating oil system capacity of any K19 engine.

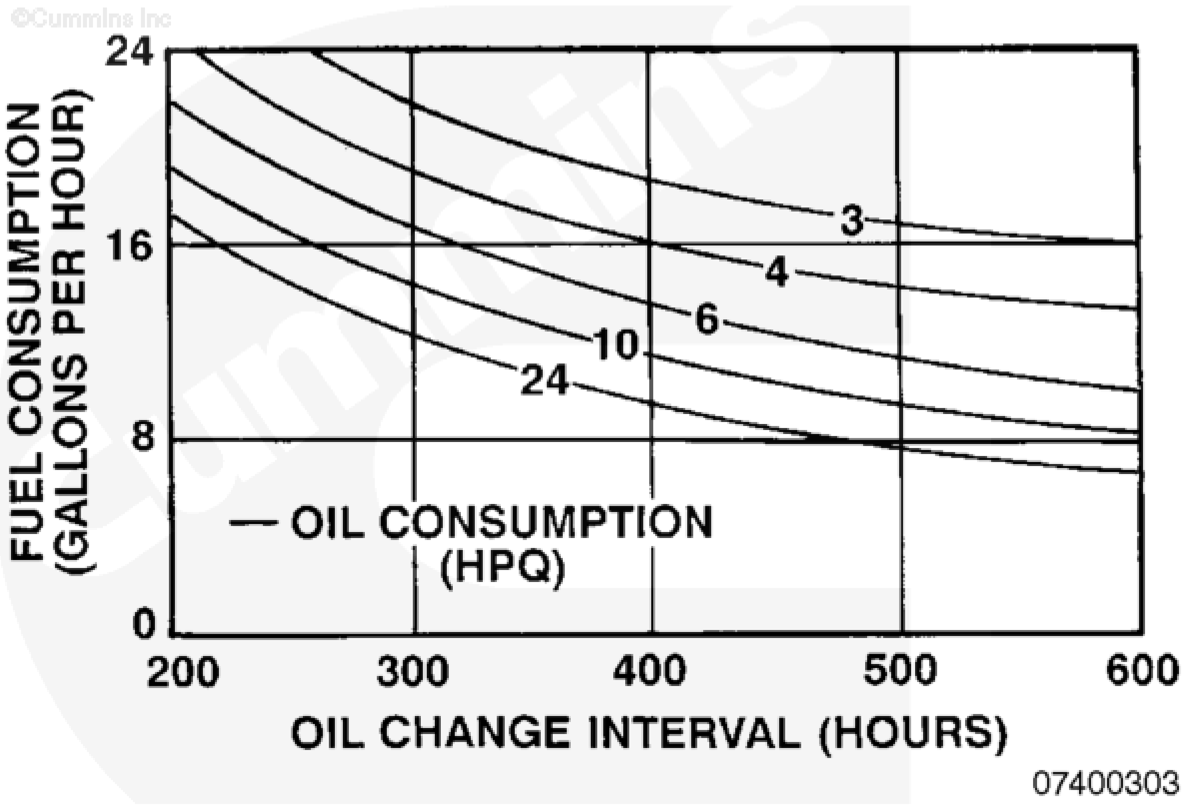
Chart Method - 42 to 53 liter [11 to 14 US gallon] capacity.



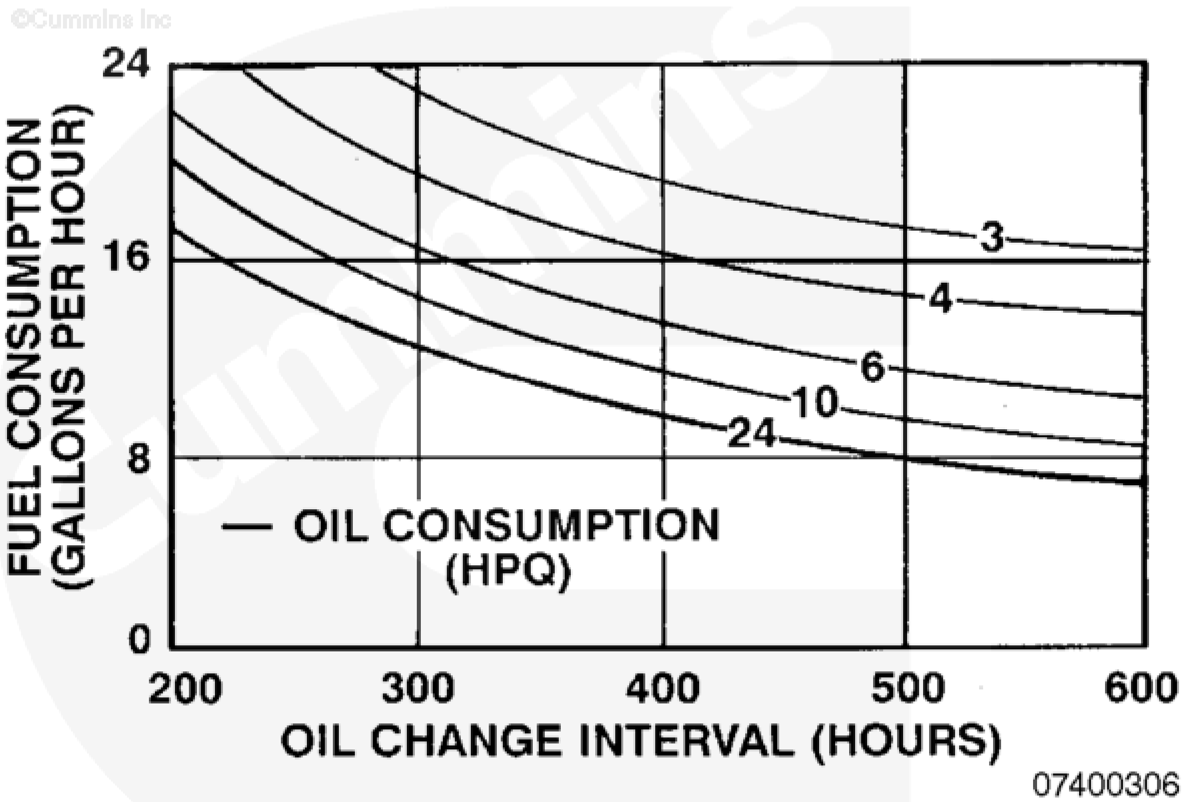
Lube System Capacity - 42 liter [11 U.S. gal]



Lube System Capacity - 45 liter [12 U.S. gal]

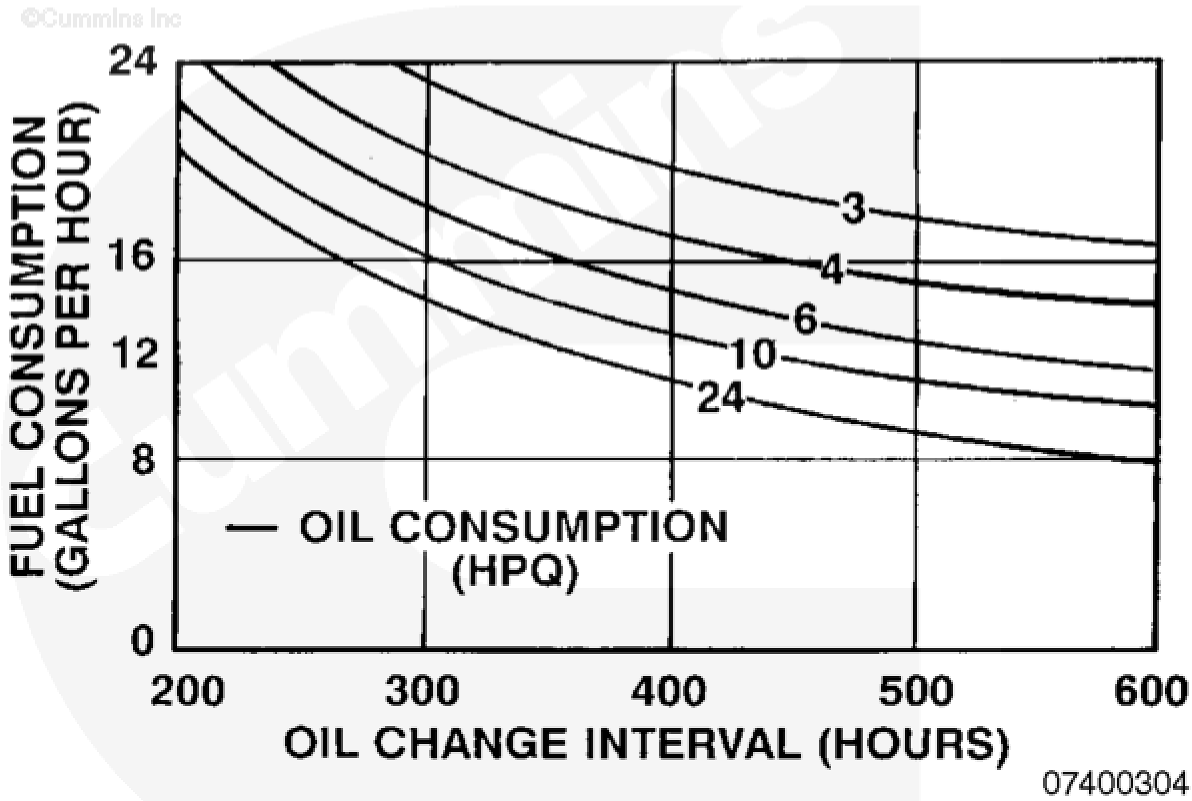


Lube System Capacity - 49 liter [13 U.S. gal]

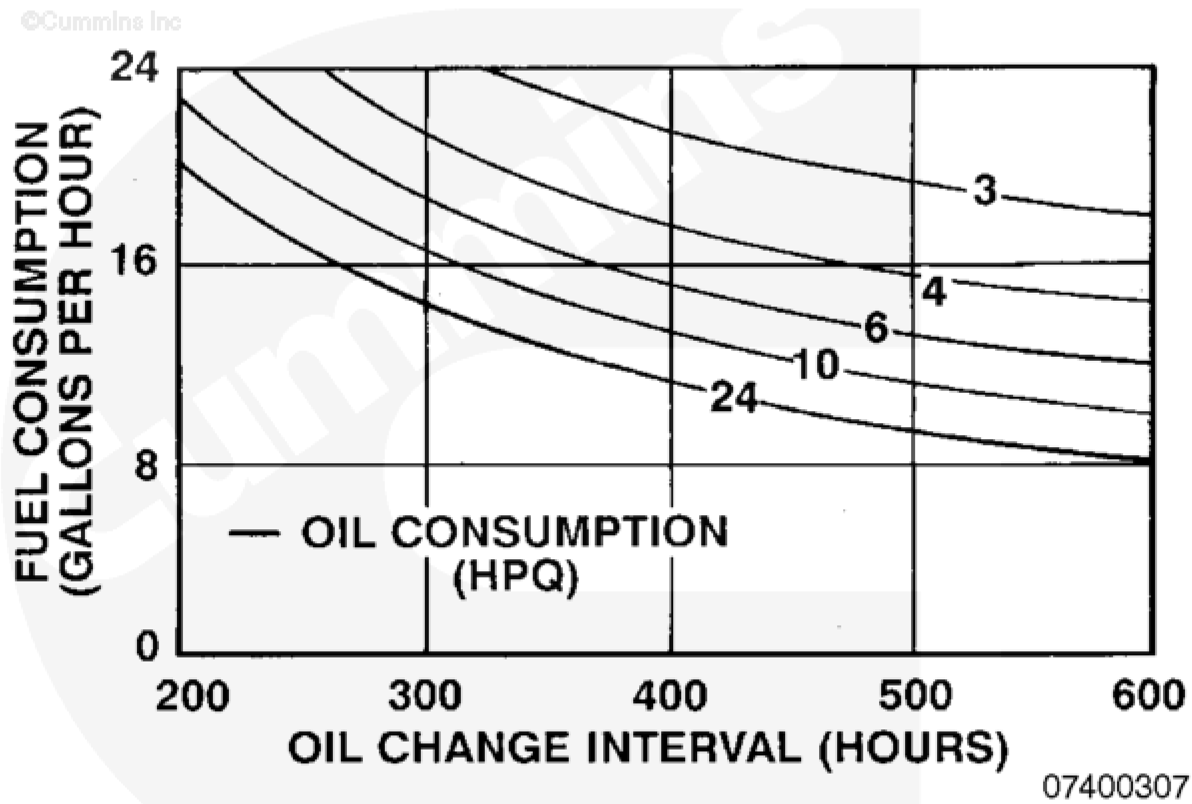


Lube System Capacity - 53 liter [14 U.S. gal]

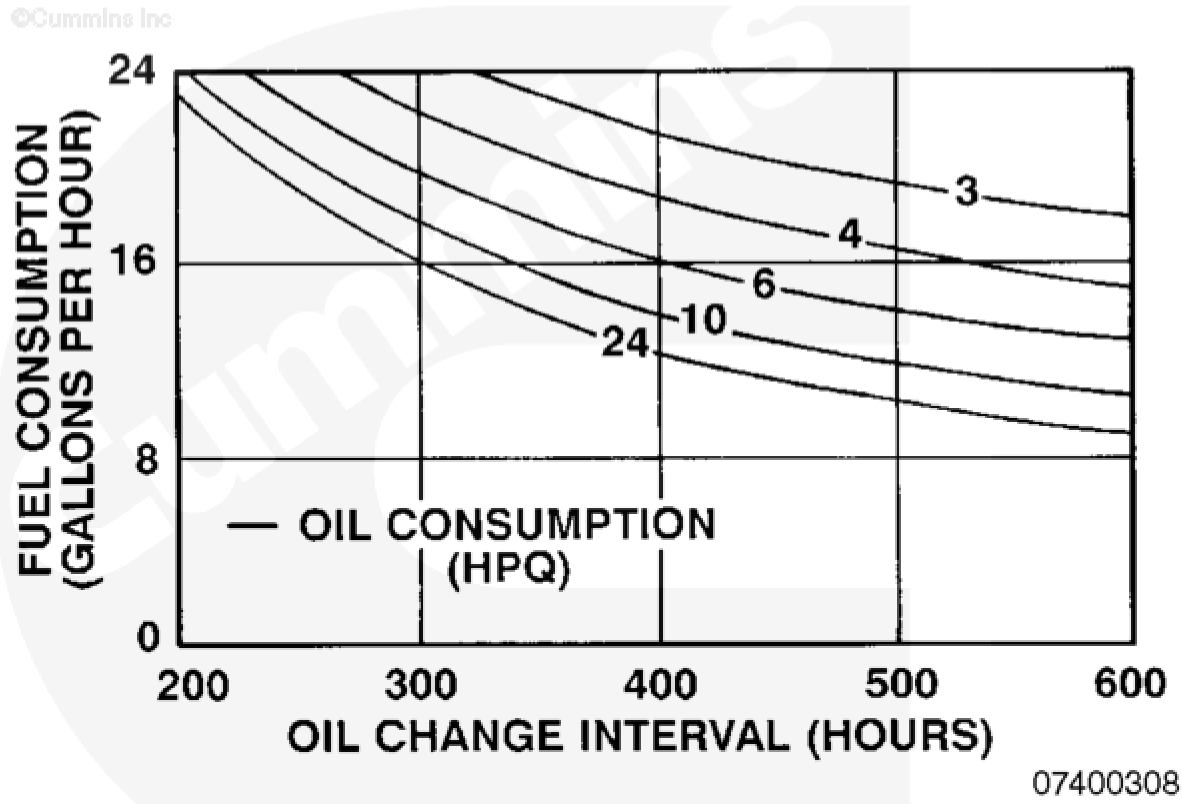
Chart Method - 57 to 68 liter [15 to 18 US gallon] capacity.



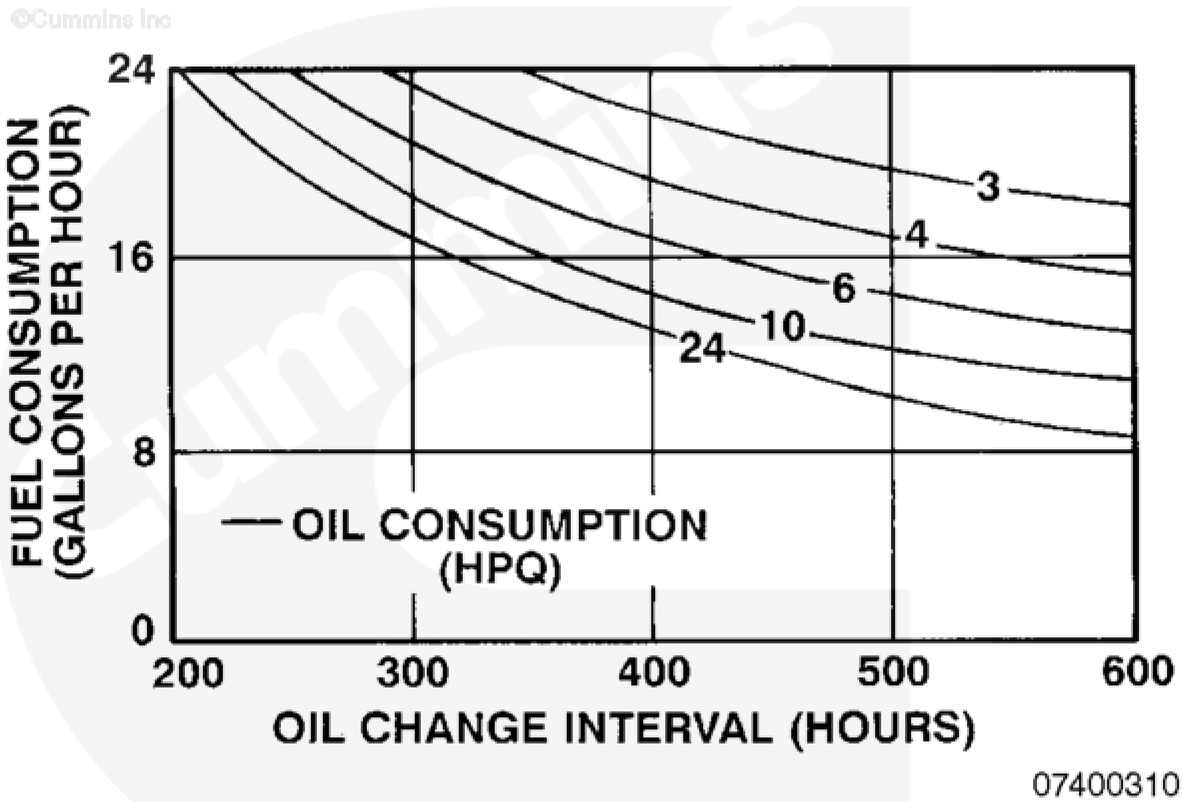
Lube System Capacity - 57 liter [15 U.S. gal]



Lube System Capacity - 61 liter [16 U.S. gal]

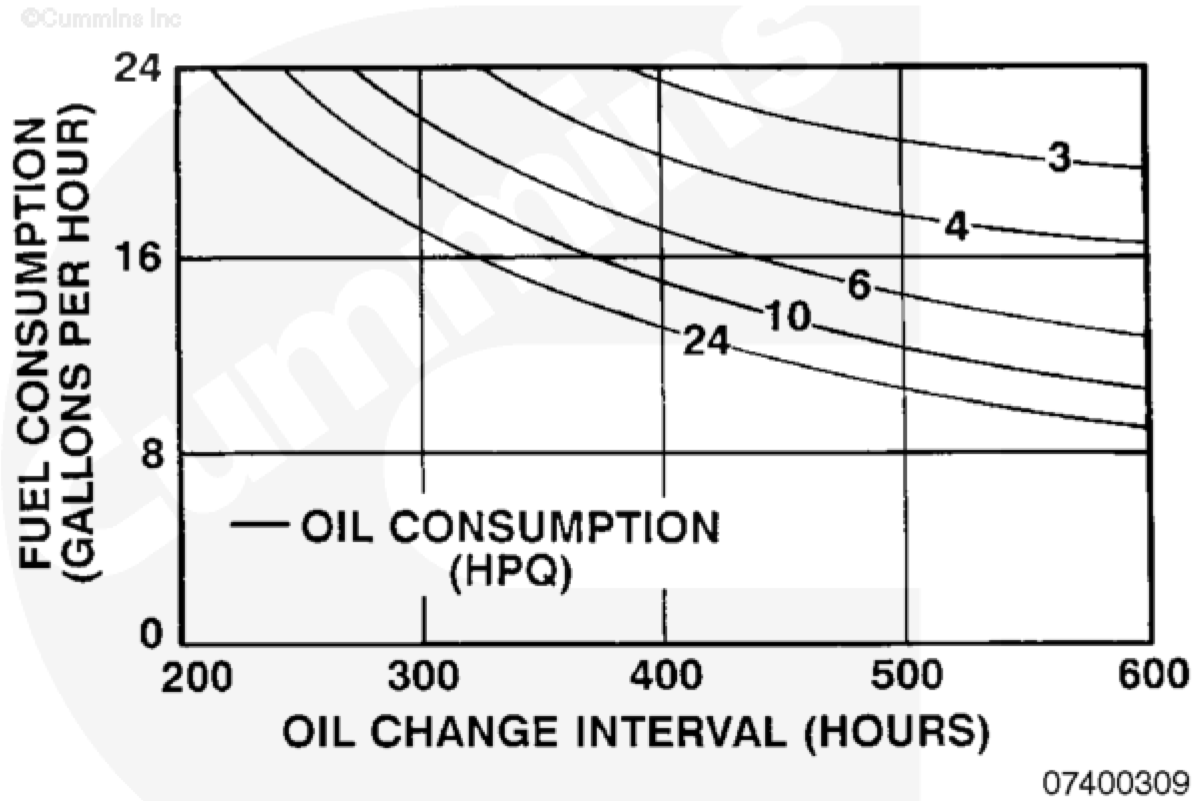


Lube System Capacity - 64 liter [17 U.S. gal]

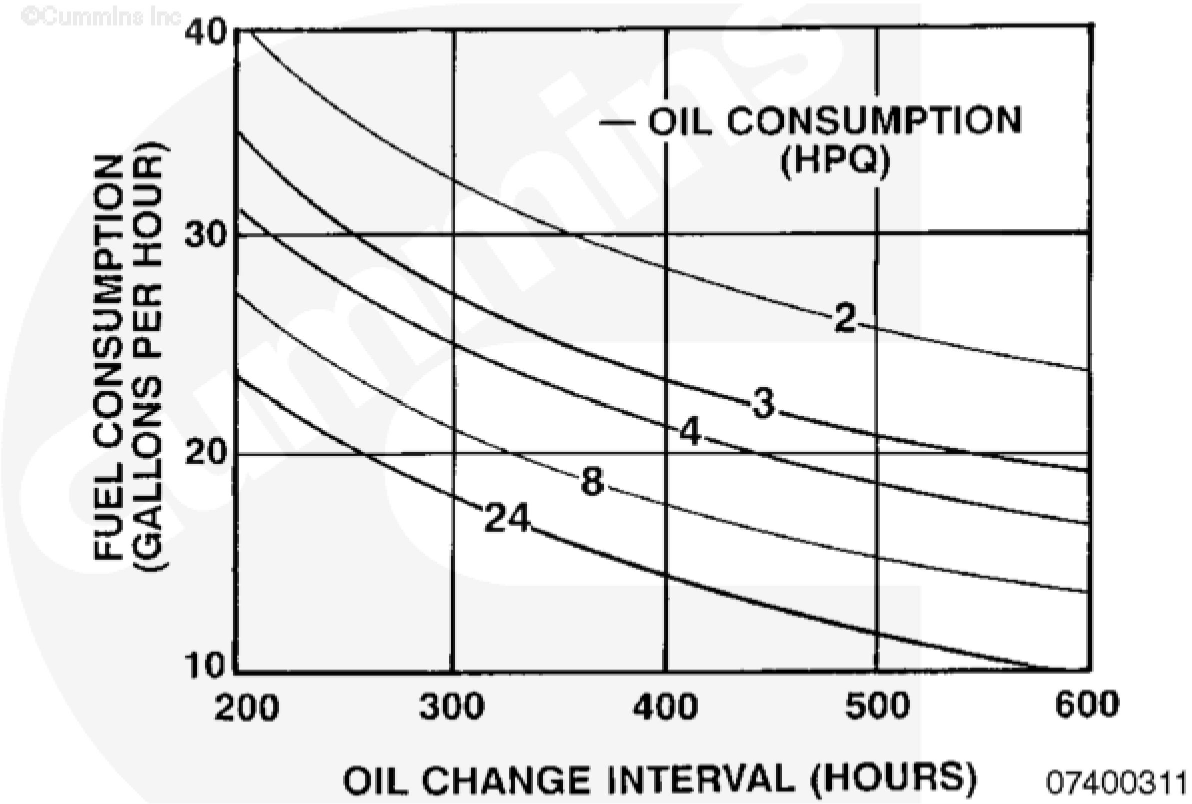


Lube System Capacity - 68 liter [18 U.S. gal]

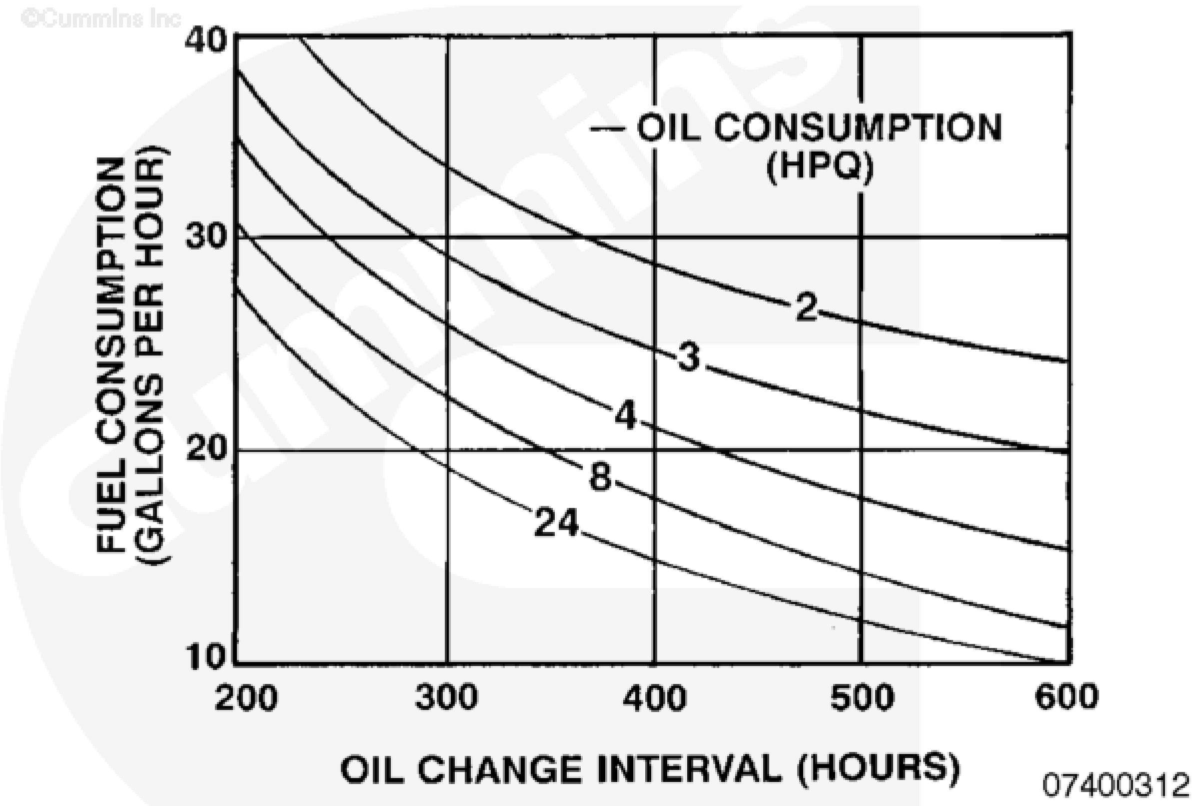
Chart Method - 72 to 83 liter [19 to 22 US gallon] capacity.



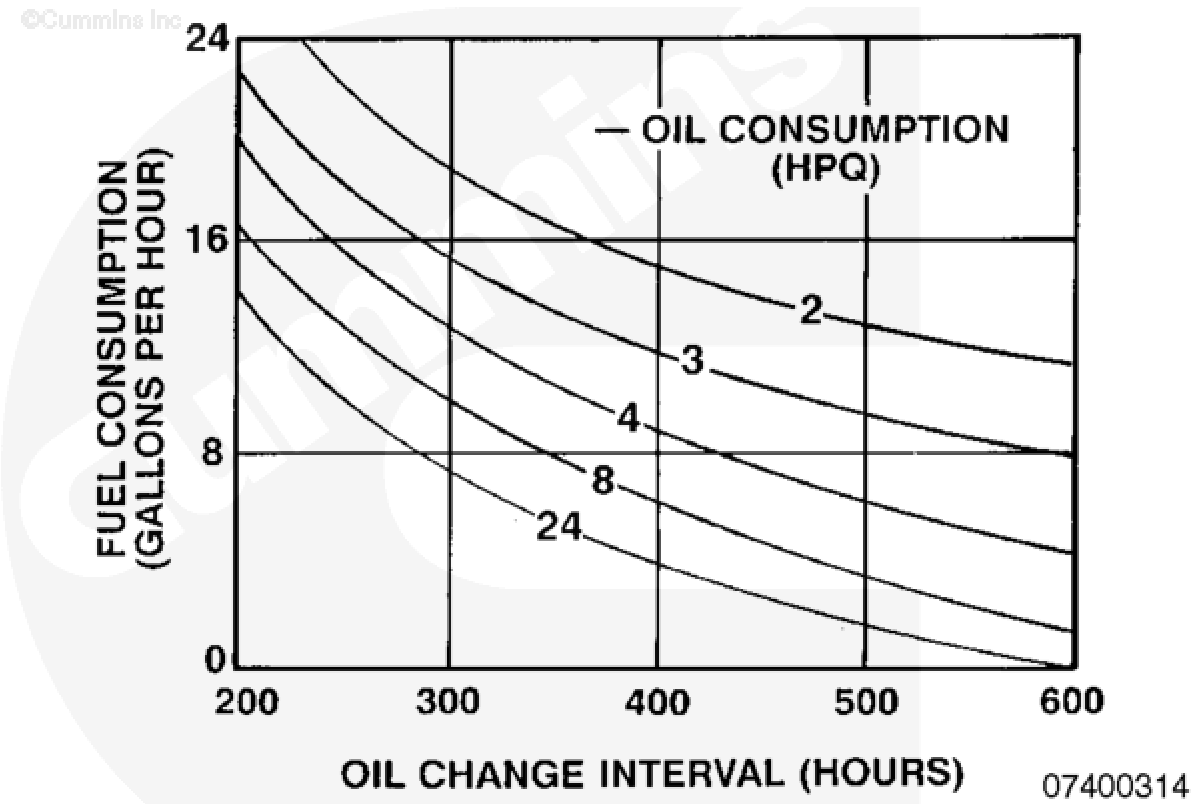
Lube System Capacity - 72 liter [19 U.S. gal]



Lube System Capacity - 76 liter [20 U.S. gal]

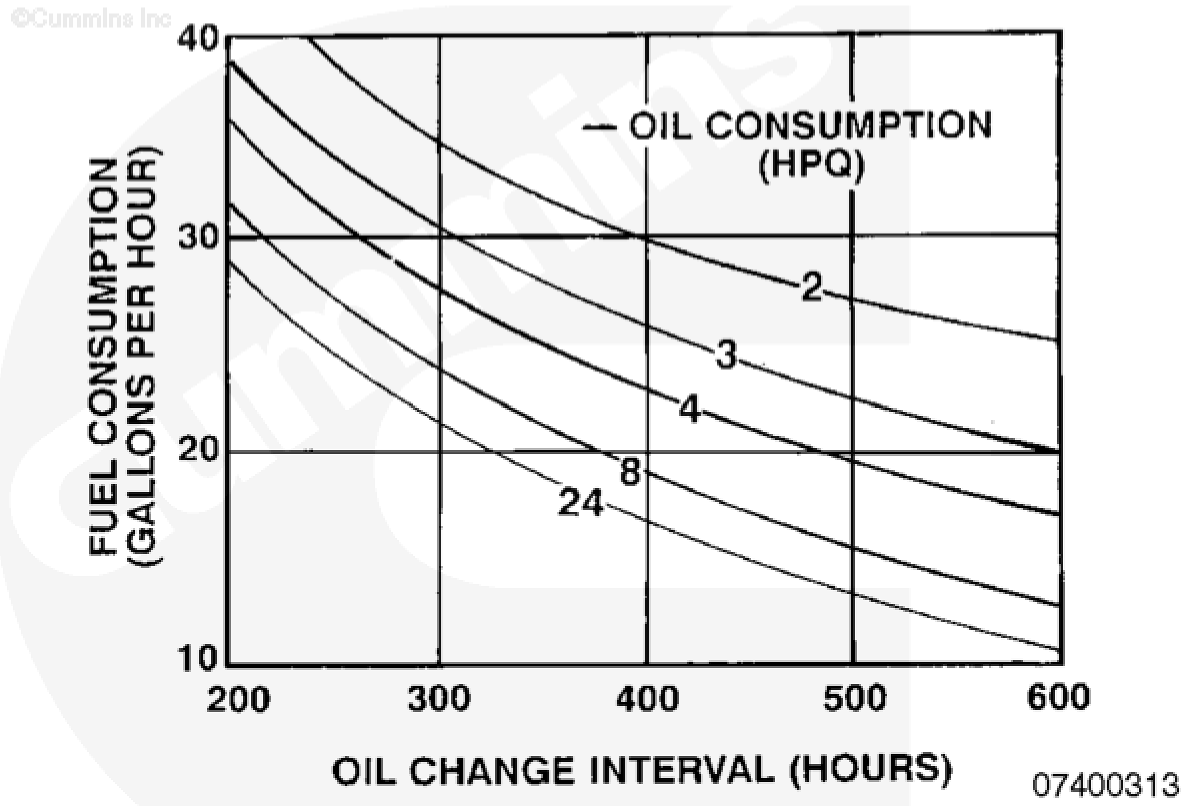


Lube System Capacity - 79 liter [21 U.S. gal]

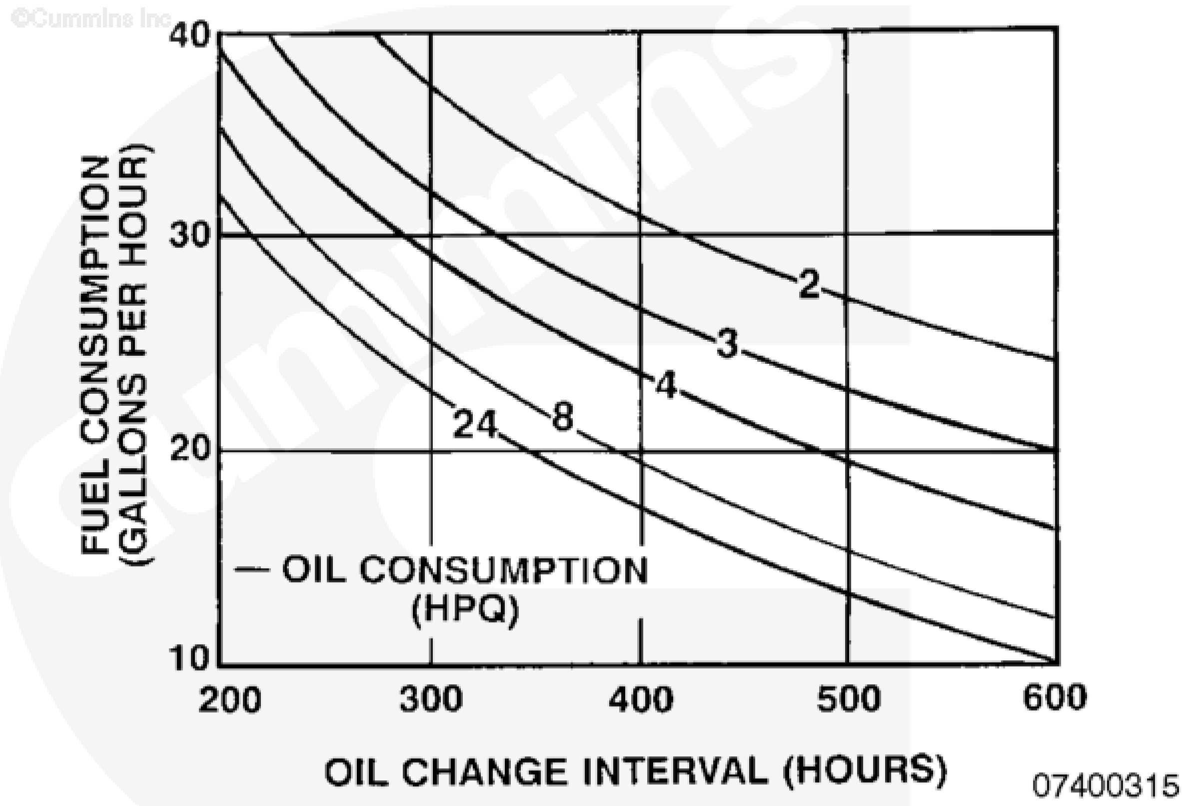


Lube System Capacity - 83 liter [22 U.S. gal]

Chart Method - 87 to 91 liter [23 to 24 US gallon] capacity.



Lube System Capacity - 87 liter [23 U.S. gal]



Lube System Capacity - 91 liter [24 U.S. gal]

Continue to Fixed Mileage or Hours.

Fixed Mileage or Hours (All Applications)

Cummins Inc. recommends an oil change interval for all K19 engine applications (except for generator drives) of 250 hours or 6 months, whichever occurs first, or an oil change interval based on the above Chart Method. The generator drive recommended fixed hour oil change interval is 250 hours or 12 months, whichever occurs first.

Last Modified: 22-Oct-2013